CRYPTOGRAPHIC AND SECURITY ANALYSIS OF THE CHC SECURE FILE MANAGEMENT SYSTEM

Project Report Submitted by

NITHIN K R

NNM24CB503

UNDER THE GUIDANCE OF

Mr. S Shyam Kumar

Assistant Professor Gd-I

Dept of Cybersecurity, NMAMIT, Nitte

in partial fulfillment of the requirements for the award of the Degree of

Bachelor of Engineering

in

CYBER SECURITY

from

Nitte (Deemed to be University)

NOV 2025

Department of Cyber Security

CERTIFICATE

Certified that the project entitled ***"CRYPTOGRAPHIC AND SECURITY ANALYSIS OF THE CHC SECURE FILE MANAGEMENT SYSTEM"*** *carried out* ***by MR. NITHIN K R , NNM24CB503****, a bonafide student of* ***NMAM Institute of Technology****, Nitte in partial fulfillment for the award of Bachelor of Engineering* ***in Cyber security*** *of the Nitte (Deemed to be University) during the year* ***2025-26****. It is certified that all corrections / suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.*

Name & Signature of Guide(s)	Name & Signature of HOD	Signature of the Principal

External Viva

Name of the Examiners	Signature with Date
1. ______________________	______________________
2. ______________________	______________________

ACKNOWLEDGEMENT

I believe that my project will be complete only after I thank the people who have contributed to make this project successful.

First and foremost, my sincere thanks to our beloved principal, Dr. Niranjan N. Chiplunkar for giving us an opportunity to carry out my project work at our college and providing us with all the needed facilities.

I express my deep sense of gratitude and indebtedness to my guide Mr. S Shyam Kumar, Assistant Professor Go-I, Dept. of Cyber security NMAMIT, Nitte for her inspiring guidance, constant encouragement, support and suggestions for improvement during the course for our project.

I acknowledge the support and valuable inputs given by Dr. Roshan Fernandes, Head of the Department, Cyber security, NMAMIT, Nitte.

Finally, I thank the staff members of the Department of Cybersecurity and our parents and all our friends for their honest opinions and suggestions throughout the course of our project.

NITHIN K R

NNM24CB503

ABSTRACT

This project presents a comprehensive, three-pillar research framework for the rigorous security analysis of the **CHC Secure File Management System**. The system integrates custom cryptography (CHC Algorithm) with a simulated block chain for metadata integrity and a Flask application for access control. The primary objective is to move beyond theoretical claims to provide quantified, empirical evidence of the system's security, performance, and robustness.

The analysis is structured across: **Pillar A: Cryptographic Novelty** (validating seed entropy, forward security, and performance against AES-256-GCM); **Pillar B: Block chain Integration** (testing data immutability, tamper detection, and auditability); and **Pillar C: System Security** (analysing key management, access control enforcement, and web vulnerabilities via OWASP Top 10 assessment). The successful execution of this framework will produce an academic-grade security report that either validates the CHC system's claims or identifies specific, actionable areas for cryptographic hardening and architectural improvement.

TABLE OF CONTENTS

LIST OF FIGURES